SAI NITHIN REDDY KASIREDDY

AI/ML ENGINEER | GENERATIVE AI | LLMs | RAG | AGENTIC AI | MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with nearly 5 years of experience designing, developing, and deploying production-grade Machine Learning, Generative AI, LLM, and Agentic AI solutions across AWS and Microsoft Azure. Experienced in building enterprise RAG platforms, AI agent workflows, LLM-powered applications, predictive ML systems, scalable data pipelines, and MLOps solutions using Python, LangChain, LangGraph, Azure OpenAI, AWS Bedrock, FastAPI, Databricks, Docker, Kubernetes, and MLflow. Proven ability to integrate structured and unstructured enterprise data, develop cloud-native AI services, automate business processes, and deliver measurable improvements in accuracy, scalability, productivity, deployment efficiency, and operational performance.

TECHNICAL SKILLS

Programming: Python, SQL, Java, Bash

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), AI Agents, Agentic AI, Predictive Modeling, Classification, Forecasting, Feature Engineering

Generative AI & LLM: Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Orchestration, Multi-Agent Workflows, OpenAI GPT-4/4o APIs, Azure OpenAI, AWS Bedrock, Hugging Face Transformers

AI/ML Frameworks: LangChain, LangGraph, LlamaIndex, PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, OpenCV

Vector Databases & Search: Pinecone, ChromaDB, FAISS, Vector Embeddings, Semantic Search

Cloud & AI Platforms: AWS (SageMaker, Bedrock, Lambda, ECS, S3), Microsoft Azure, Azure AI Studio, Azure Machine Learning

Data Engineering: Apache Spark, PySpark, Databricks, Delta Lake, Apache Kafka, Apache Airflow, Snowflake

MLOps & DevOps: MLflow, Docker, Kubernetes, Terraform, Git, GitHub Actions, Jenkins, CI/CD, Model Deployment, Model Monitoring, Model Versioning

Backend & APIs: FastAPI, REST APIs, JSON

Databases: Snowflake, PostgreSQL, SQL Server, MySQL

Tools & Methodologies: Power BI, Tableau, JIRA, Confluence, VS Code, Jupyter Notebook, Linux, GitHub, Agile/Scrum, SDLC, Machine Learning Lifecycle, MLOps

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Palantir Technologies, USA **Aug 2025 – Present**

- Designed and deployed production-grade Generative AI applications using Python, LangChain, LangGraph, Azure OpenAI, FastAPI, and LLM orchestration, reducing manual knowledge retrieval by 65% while improving response accuracy and enterprise productivity.
- Built scalable Retrieval-Augmented Generation (RAG) pipelines using Pinecone, ChromaDB, FAISS, and Snowflake to enable semantic search across 12M+ enterprise documents, achieving 92% retrieval accuracy.
- Engineered LangGraph-based AI agent workflows for document analysis, compliance validation, report generation, and business process execution, reducing analyst turnaround time by 55%.
- Implemented enterprise MLOps workflows using MLflow, Apache Airflow, Docker, Kubernetes, GitHub Actions, and Terraform for automated model deployment, monitoring, and versioning, reducing deployment time by 70%.
- Developed distributed AI data-processing pipelines using Apache Spark, Databricks, Kafka, and Delta Lake, processing 1.5TB+ of enterprise data daily and improving pipeline throughput by 45%.

- Integrated structured and unstructured enterprise data sources into cloud-native AI services and collaborated with engineering, product, and security teams to maintain 99.9% platform availability and accelerate enterprise AI adoption.
- Supported scalable AI application deployment across AWS and Azure, integrating APIs, data pipelines, LLM services, vector search, and containerized infrastructure into production workflows.

Data Scientist | Delta Electronics Inc., India **Jan 2021 – Aug 2024**

- Developed and deployed machine learning models for predictive maintenance and demand forecasting using Python, TensorFlow, Scikit-learn, XGBoost, and Pandas, improving prediction accuracy by 27% and reducing equipment downtime by 30%.
- Designed scalable ETL and feature-engineering pipelines using Azure Data Factory, Azure Machine Learning, Databricks, PySpark, SQL, and Delta Lake, processing 600GB+ of manufacturing data daily.
- Built NLP-based document classification and intelligent search solutions using Hugging Face Transformers, OpenAI APIs, and LangChain, reducing manual document-processing effort by 50%.
- Implemented ML lifecycle management using MLflow, Git, Docker, Azure DevOps, and CI/CD pipelines, reducing model deployment cycles by 40% while improving reproducibility and governance.
- Developed interactive Power BI and Tableau dashboards delivering operational insights to 200+ business stakeholders, reducing reporting preparation time by 60%.
- Collaborated with business and technical stakeholders to translate manufacturing requirements into predictive analytics, machine learning, NLP, and data-driven solutions.

ACADEMIC PROJECTS

Enterprise Document Intelligence using Retrieval-Augmented Generation (RAG)

(Python | LangChain | OpenAI GPT-4 | ChromaDB | FastAPI | Streamlit)

- Developed a RAG-based document intelligence application enabling semantic search and question answering over enterprise documents using LLMs and vector embeddings.
- Optimized document chunking, embedding strategies, and prompt engineering to improve document retrieval accuracy by 35%.
- Built an API-driven workflow using FastAPI, integrating vector search with LLM-based response generation.

Predictive Customer Churn Analytics

(Python | Scikit-learn | XGBoost | Pandas | SQL | Power BI)

- Built an end-to-end machine learning pipeline for customer churn prediction using feature engineering and classification models.
- Achieved 91% model accuracy and developed interactive Power BI dashboards to identify high-risk customers and support data-driven retention strategies.

EDUCATION

Master of Professional Studies (MPS) in Data Science ***May 2026***

University of Maryland, Baltimore County (UMBC), Catonsville, MD

Bachelor of Technology (B.Tech.) in Electronics and Communication Engineering ***June 2023***

Vellore Institute of Technology (VIT), Vellore, India

CERTIFICATIONS

Machine Learning with Python Professional Certificate by Anaconda — LinkedIn Learning / Anaconda | Aug 2026

Getting Started with Generative AI — IBM SkillsBuild

Artificial Intelligence Fundamentals — IBM SkillsBuild